

Further sardine length-weight regressions for survey and commercial lengths

by

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1 Summary

A TG of the PWG has revisited the weight to length relationships used in the management of the South African sardine resource. This document contains additional regression results for weight to length, including those applicable when additional data (more years) for the recruit survey are included. Results are specific to surveys or commercial catches, however within those categories, year specific or combined, and, coast specific or combined results are produced and need to be considered by the PWG. This revised document corrects an error in the plotting of Figure 3.

Final decisions are required for the regressions to be used for the recruit surveys, the November biomass surveys and for the West and South Coast commercial data.

2 Background

In 2020, based on results presented in **FISHERIES/2020/MAR/SWG-PEL/22REV¹**, a TG of the PWG decided to adopt the following basis for calculating sardine weight (W) from sardine length (L):

1. **Commercial catch, South Coast** – the parameters from a linear regression of $\ln(W)$ against $\ln(L)$ where W is in units of grams and L is in units of millimetres. Data: South Coast samples from commercial catches as supplied to OLSPS Marine by Carl van der Lingen, 2019. Data excluded: $L < 150$ mm.
2. **Commercial catch, West Coast** – the parameters from a linear regression of $\ln(W)$ against $\ln(L)$ where W is in units of grams and L is in units of millimetres. Data: West Coast samples from commercial catches as supplied to OLSPS Marine by Carl van der Lingen, 2019. No data exclusions.

¹ OLSPS Marine. 2020. Updated length weight regression parameters for sardine. 19 March 2020 Revised 12 June 2020. FISHERIES/2020/MAR/SWG-PEL/22REV

3. **Surveys on both the West and South Coasts:** -- the parameters from a linear regression of $\ln(W)$ against $\ln(L)$ where W is in units of grams and L is in units of millimetres. Data: West Coast and South Coast samples combined from the surveys as supplied to OLSPS Marine by Carl van der Lingen, 2019. No data exclusions.

The parameter estimates associated with the above are given in Table 1.

Table 1. Estimates of the parameters of the power relationship relating body weight W (grams) to caudal length L (millimetres) in the relationship $W = aL^b$. For surveys the same relationship is application for both West and South Coasts whereas for commercial catches there are different parameters for the West and South Coasts, as reported in FISHERIES/2020/MAR/SWG-PEL/22REV.

Commercial or Survey	Coast	L expressed in millimetres and W in grams		
		b	lna	a
Commercial	South Coast	2.463739	-8.375636	0.00023041000
Commercial	West Coast	3.066305	-11.616146	0.00000901930
Survey	West Coast and South Coast Combined	3.140026	-12.077216	0.00000568760

As part of the revision of the pelagic OMP, specifically the sardine component thereof, during 2023, further work was requested, exploring whether to allow for differences in the W to L regression for sardine by considering possible year, coast and survey specific relationships (the latter either mid-year recruitment or November adult biomass surveys).

As a result and at the request of a TG of the PWG, further estimates were produced to facilitate decision making in this regard. These results and associated data and methods are presented here to facilitate PWG deliberations on this matter.

3 Data and Methods

The data used here are those described in FISHERIES/2020/MAR/SWG-PEL/22REV, but complemented by additional fish length and fish weight data for recruit surveys for years 2019 to 2023. The numbers of fish lengths and weights available for analysis are presented in Table 2.

Table 2. The number of fish length and weight records available to the analyses reported on in this document. Note that the new data for the recruit survey indicated was not available to the work reported in FISHERIES/2020/MAR/SWG-PEL/22REV.

	Commercial Data		Survey Data	
Year	West Coast	South Coast	Recruit	November
2000	3408	696		
2001	3521	1024		
2002	5147	1447		
2003	3650	1564		
2004	2751	2425		
2005	1152	3409	2058	814
2006	1225	1642	2723	772
2007	1520	899	984	822
2008	891	722	482	506
2009	948	793	996	732
2010	1150	274	1107	1698
2011	1349	327	1094	1648
2012	700	540	1491	770
2013	225	271	659	744
2014	884	965	1055	1107
2015	1589	374	2234	1058
2016	1811	1744	942	1207
2017	1248	1016	1547	932
2018	1400	1256	New Data Follows	922
2019			1251	
2020			751	
2021			1784	
2022			753	
2023			2041	

The methods used when years are combined were to carry out linear regressions of $\ln W$ vs $\ln L$, for W in grams and L now in centimetres, i.e.

$$\ln W = \mu + \alpha \ln L$$

When allowing for year specific differences, a GLM with $\ln W$ as the target variable, and the predictors year y and $\ln L$ as well as the two-way interaction between year y and $\ln L$, was used, viz.

$$\ln W = \mu + \gamma_y + \alpha \ln L + \beta_y \ln L$$

Where the notation definitions are:

$\ln W$; the target variable

μ ; the model intercept

γ_y ; the year effect parameters

α ; the slope parameter of the covariate $\ln L$

β_y ; slope parameters of the covariate $\ln L$ arising from including the two way interaction term $y \times \ln L$.

The conventional formula relating weight W to length L is

$$W = aL^b$$

For the year independent formulation used here, this is:

$$W = e^{\mu} L^{\alpha},$$

so that for the conventional application $a = e^{\mu}$ and $b = \alpha$. For a year specific case coupled with an extension of the conventional, the formula would be:

$$W = a_y L^{b_y}.$$

For the formulation used here this corresponds to

$$W = e^{\mu + \gamma_y} L^{\alpha + \beta_y}$$

i.e. $a_y = e^{\mu + \gamma_y}$ and $b_y = \alpha + \beta_y$.

4 Results

For the commercial data only year combined results were run. Year specific estimates were however explored when using the survey data.

4.1 Commercial results

Table 3. Parameter estimates and standard errors for the commercial data.

West Coast					
All L		Estimate	Standard Error		
	μ	-4.556	0.014	a	0.0105040
	α	3.066	0.005	b	3.0660000
South Coast					
All L		Estimate	Standard Error		
	μ	-3.015	0.029	a	0.0490458
	α	2.571	0.010	b	2.5710000
South Coast					
L >= 15		Estimate	Standard Error		
	μ	-2.703	0.030	a	0.0670042
	α	2.464	0.010	b	2.4640000

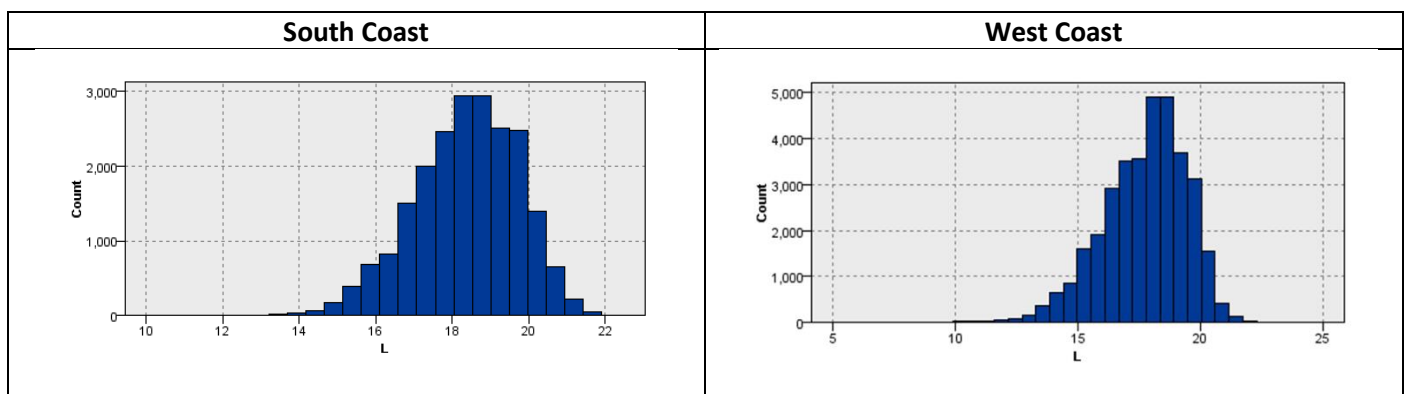


Figure 1. Length frequency plots for the commercial data available for the analyses, summed up over all years.

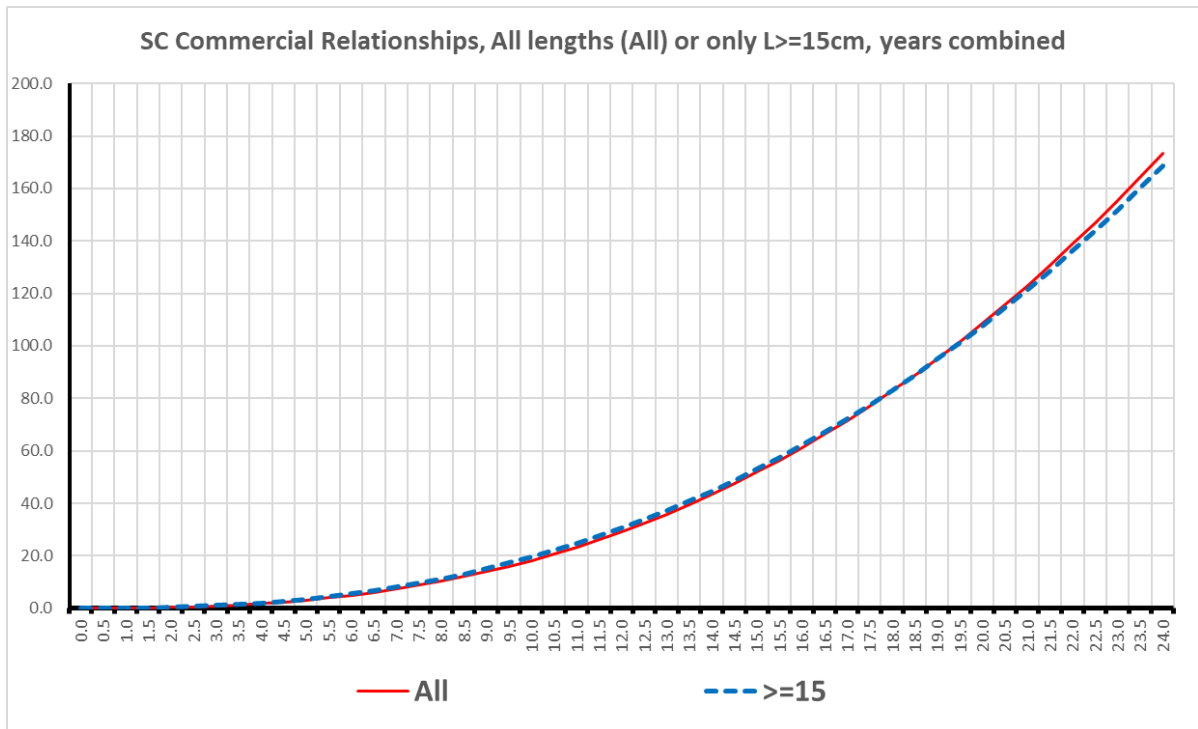


Figure 2. Plot of the relationship W to L for the South Coast commercial data, comparing using all L or only L larger than 15 cm.

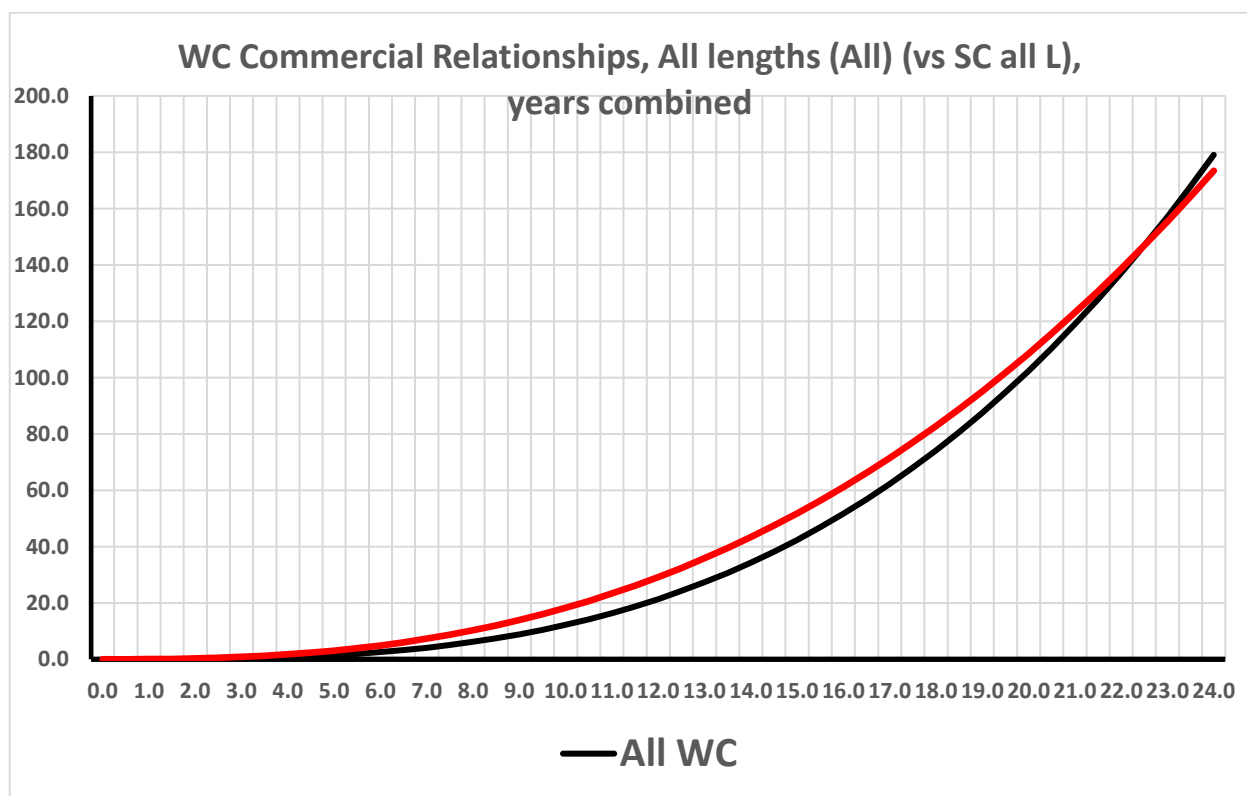


Figure 3. Plot of the relationship W to L for the West Coast commercial data, compared to the South Coast relationship with all L .

4.2 Survey results

4.2.1 Year Specific

Table 4. Parameter estimates for the Recruit and November survey datasets, for the case where year specific estimates are permitted, and West and South Coast datasets are combined.

		Recruit Survey			November Survey		
		Estimate	s.e.	p	Estimate	s.e.	p
μ		-4.777	0.024	0.000	-4.699	0.022	0.000
	2005	0.209	0.076	0.006	0.132	0.069	0.056
γ_y	2006	-0.392	0.055	0.000	-0.470	0.050	0.000
	2007	-0.057	0.044	0.190	-0.135	0.040	0.001
	2008	-0.121	0.072	0.091	-0.198	0.065	0.002
	2009	-0.281	0.081	0.001	-0.359	0.074	0.000
	2010	-0.215	0.048	0.000	-0.293	0.043	0.000
	2011	-0.194	0.070	0.006	-0.271	0.064	0.000
	2012	0.552	0.042	0.000	-0.628	0.059	0.000
	2013	-0.192	0.047	0.000	-0.046	0.074	0.533
	2014	-0.042	0.044	0.345	-0.620	0.031	0.000
	2015	-0.330	0.042	0.000	-0.703	0.048	0.000
	2016	0.083	0.051	0.107	-0.113	0.036	0.002
	2017	0.217	0.035	0.000	-0.270	0.039	0.000
	2018				0.000	.	.
	2019	0.041	0.032	0.207			
	2020	-0.137	0.046	0.003			
	2021	0.018	0.033	0.572			
	2022	-0.038	0.035	0.281			
	2023	0.000	.	.			
β_y	2005	0.184	0.026	0.000	3.051	0.023	0.000
	2006	0.412	0.019	0.000	3.279	0.017	0.000
	2007	0.272	0.014	0.000	3.139	0.013	0.000
	2008	0.299	0.026	0.000	3.166	0.024	0.000
	2009	0.355	0.028	0.000	3.222	0.026	0.000
	2010	0.329	0.016	0.000	3.196	0.014	0.000
	2011	0.305	0.023	0.000	3.172	0.021	0.000
	2012	-0.008	0.014	0.545	3.332	0.021	0.000
	2013	0.324	0.015	0.000	3.113	0.026	0.000
	2014	0.257	0.014	0.000	3.332	0.008	0.000
	2015	0.368	0.013	0.000	3.340	0.016	0.000
	2016	0.196	0.017	0.000	3.128	0.012	0.000
	2017	0.160	0.011	0.000	3.209	0.013	0.000
					3.065	0.009	0.000
	2019	0.228	0.009	0.000			
	2020	0.284	0.016	0.000			
	2021	0.214	0.009	0.000			
	2022	0.258	0.010	0.000			
	2023	0.232	0.010	0.000			
α		2.867			0.000	.	.

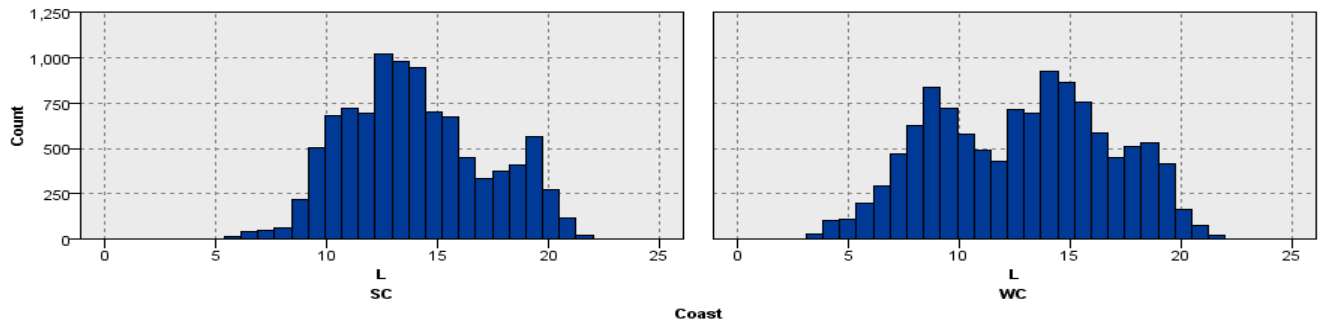


Figure 4. Length frequency distribution for the recruit survey data, South Coast LHS and West Coast RHS.

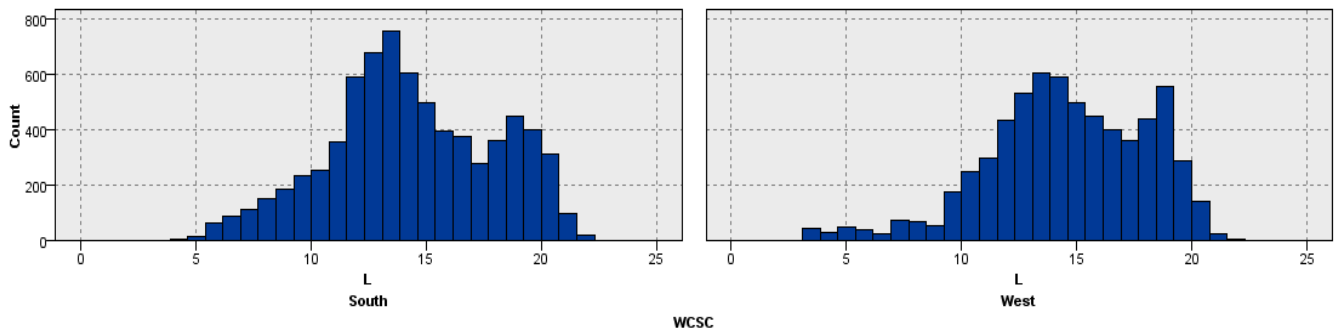


Figure 5. Length frequency distribution for the November biomass survey data, South Coast LHS and West Coast RHS.

Table 5. Parameter estimates for the Recruit and November survey datasets, for the West and South Coasts alone, and when combining the West and South Coast datasets.

WC		Estimate	Standard Error		
	μ	-4.815	0.0086	a	0.0081072
	α	3.125	0.0034	b	3.1250000
South Coast (Recruit Survey)					
SC		Estimate	Standard Error		
	μ	-4.835	0.0138	a	0.0079467
	α	3.127	0.0053	b	3.1270000
Both Coasts Combined (Recruit Survey)					
WC+SC		Estimate	Standard Error		
	μ	-4.81	0.0072	a	0.0081479
	α	3.121	0.0028	b	3.1210000
West Coast (November Survey)					
WC		Estimate	Standard Error		
	μ	-5.031	0.0129	a	0.0065323
	α	3.216	0.0049	b	3.2160000
South Coast (November Survey)					
SC		Estimate	Standard Error		
	μ	-4.861	0.0132	a	0.0077427
	α	3.146	0.005	b	3.1460000
Both Coasts Combined (November Survey)					
WC+SC		Estimate	Standard Error		
	μ	-4.946	0.009	a	0.0071118
	α	3.181	0.004	b	3.1810000

4.3 Plots for Survey Results

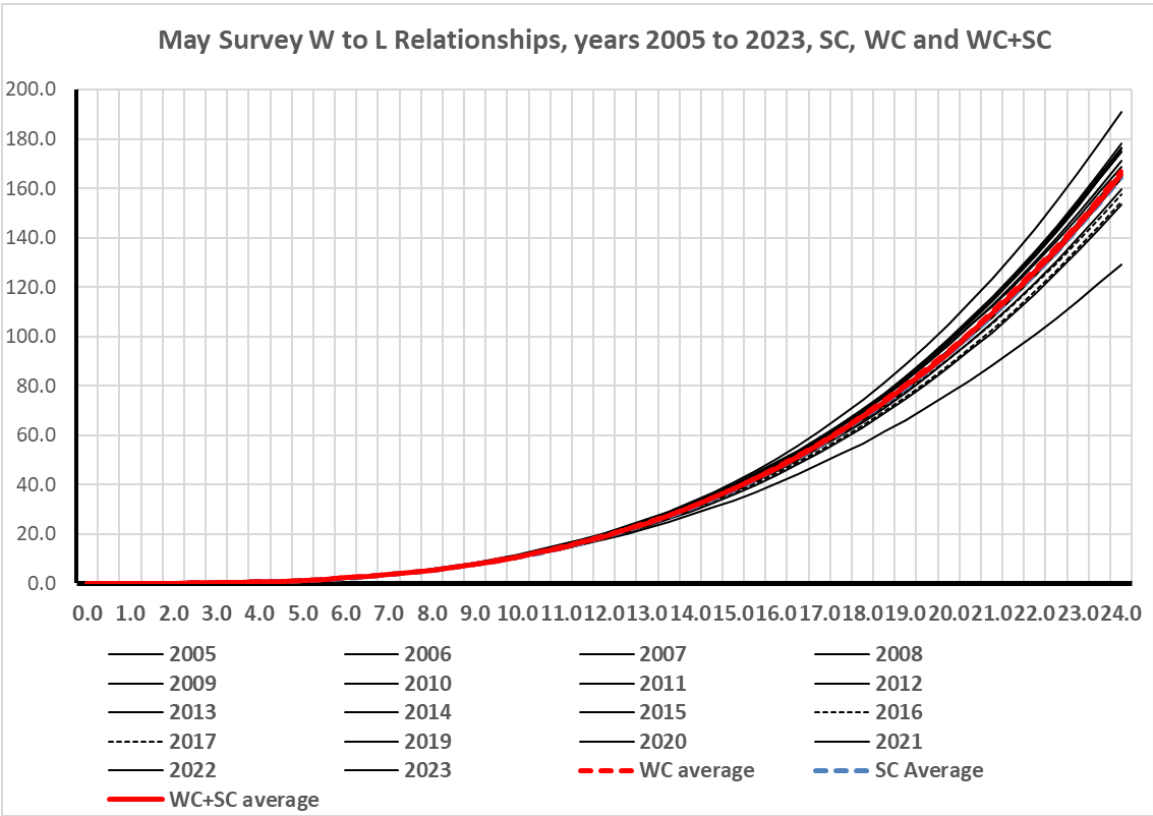


Figure 6. The fitted relationship between W (grams) and L (cm) for the recruit survey, for all years separately (but coast combined) and for the WC alone years combined, as well as the SC only and then the WC and SC combined.

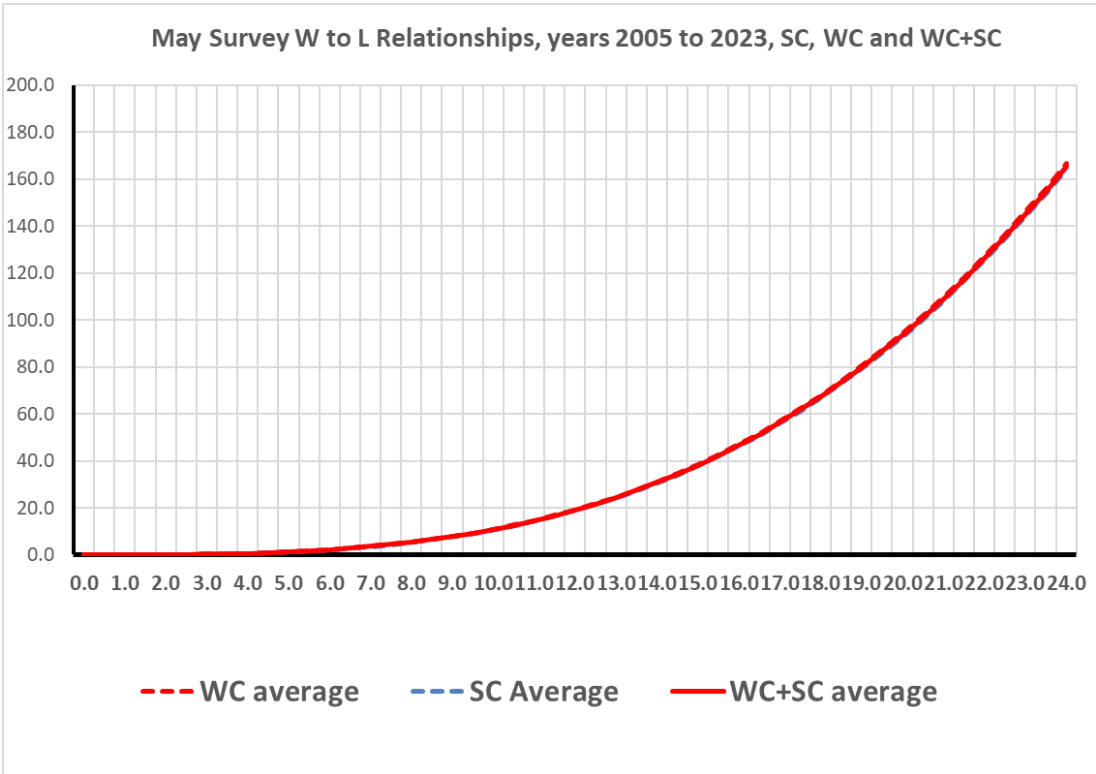


Figure 7. The fitted relationship between W (grams) and L (cm) for the recruit survey, for the WC alone years combined, as well as the SC only and then the WC and SC combined.

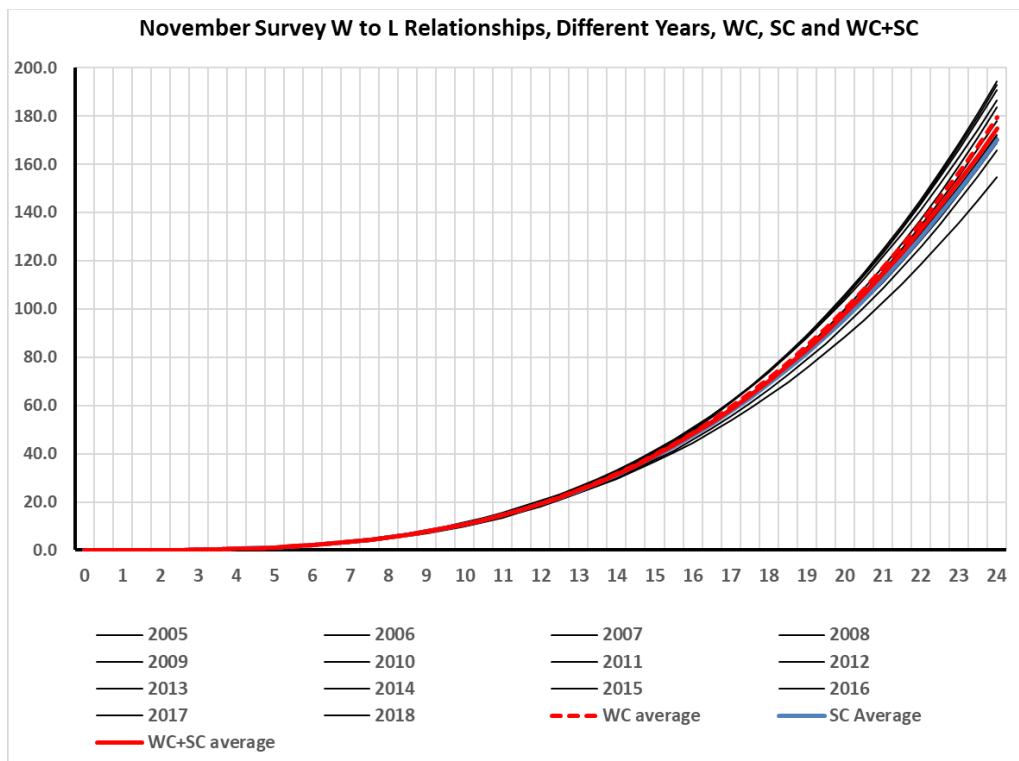


Figure 8. The fitted relationship between W (grams) and L (cm) for the November biomass survey, for all years separately (but coast combined) and for the WC alone years combined, as well as the SC only and then the WC and SC combined

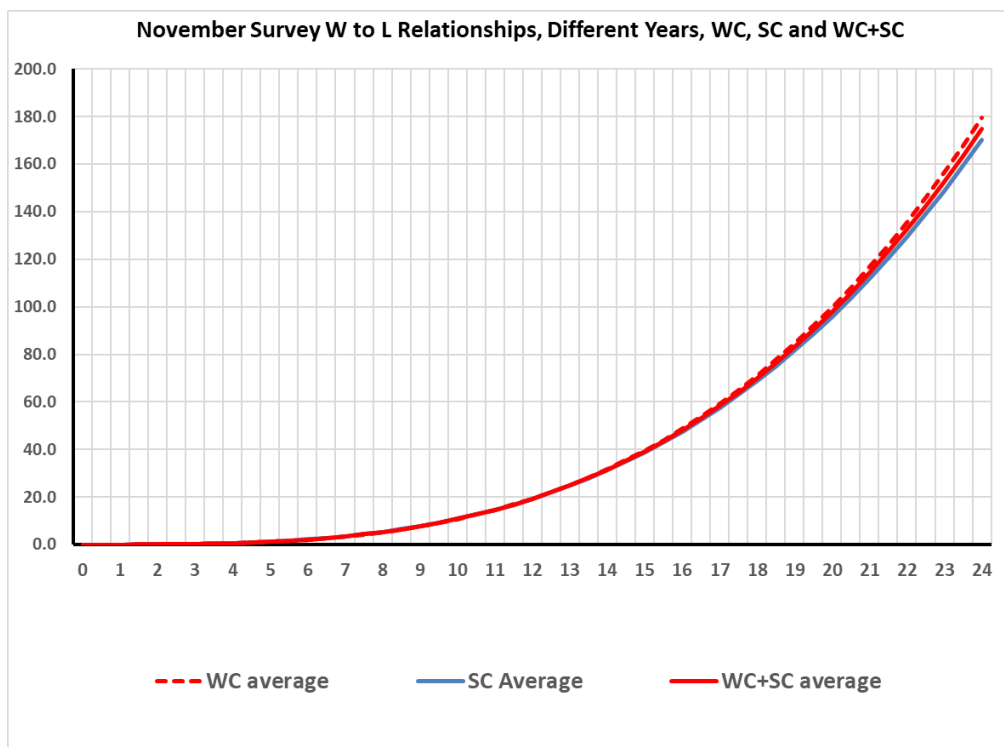


Figure 9. The fitted relationship between W (grams) and L (cm) for the November biomass survey, for the WC alone years combined, as well as the SC only and then the WC and SC combined.